Automated Weighing & Dosing Station

Concept dossier · TRSYP 3.0 Laboratory 4.0 Challenge · **Axis 2** · Team: [team name] · IEEE [student branch]

1. Which bench task are we automating, and why does it matter?

We automate **weighing and dosing a powder to a target mass** (for example 10.0 g of a material), which today is done with a spatula and a scale. The operator adds and removes powder by eye, which is slow and repetitive. It is also where labs quietly lose precision and traceability:

- a small dosing error spreads through a whole batch, and two operators never dose the same way;
- repetitive gestures cause musculoskeletal disorders, and manual steps are rarely logged in a structured way.

Our station targets the three qualities of the challenge: **dosing precision** (0.1 g, proven by measurement), **reproducibility** (the recipe, not the operator, defines the action) and **traceability** (every dose leaves a digital record). One design rule drives everything: **no timed dispensing**. Every action is decided from the measured weight.

2. What will we build, and how is it used?

A compact station with **one hopper, one auger driven by a stepper motor, and one load cell** under the receiving cup, controlled by an ESP32 and supervised from a PC interface. We chose an auger because one motor step moves a small, repeatable amount of powder (fine resolution), the powder only moves when the screw turns (clean stop, with a few reverse steps to cut off the flow), and it handles light clumping better than free-fall systems.

Use, step by step:

- **Recipe input.** The operator loads or edits a digital recipe on the PC: recipe ID, material ID, target mass (g) and tolerance (g). The format already supports several lines (one per material), although the demo uses one.
- **Sample tracking.** The operator enters a batch/sample ID. The system refuses duplicates, so each sample is unique.
- **Start validation.** The START button only becomes active when the pre-start checks are green: cup present, tare stable, hopper level sufficient, recipe valid, sample ID unused.
- **Dosing and result.** The ESP32 doses, verifies and returns the result. The small LCD shows only the key info of the current sample; the PC interface holds the full detail and history.

3. What are the main parts, and how do they connect?

Group	Parts	Role
Mechanism	Transparent conical hopper, horizontal auger in a 3D-printed tube, NEMA 17 stepper (rubber grommets), small vibration motor on the hopper wall, laser-cut MDF frame	Meters the powder; the vibration motor only prevents bridging and is switched off in the fine phase
Weighing unit	One bar load cell + HX711, light platform shaped as a steep smooth cone with a lip that fits only the cup, removable spill tray	Measures the delivered mass; the cone sheds stray powder, the lip makes the cup position repeatable
Isolation	Separate weighing base on rubber feet over a foam layer, no rigid link with the dosing tower, slack cables	Keeps motor vibration out of the weight reading
Sensors	Load cell, IR presence sensor aimed at the cup wall, ToF level sensor on top of the hopper	Weight, cup detection, powder level
Controller	ESP32 (control loop, safety), stepper driver, MOSFET module for the vibration motor	Real-time control; keeps safety even if the PC disconnects
Interface	PC dashboard over USB serial (Python), 20x4 LCD, status LED	Recipe and sample entry, checks, live view, log and history

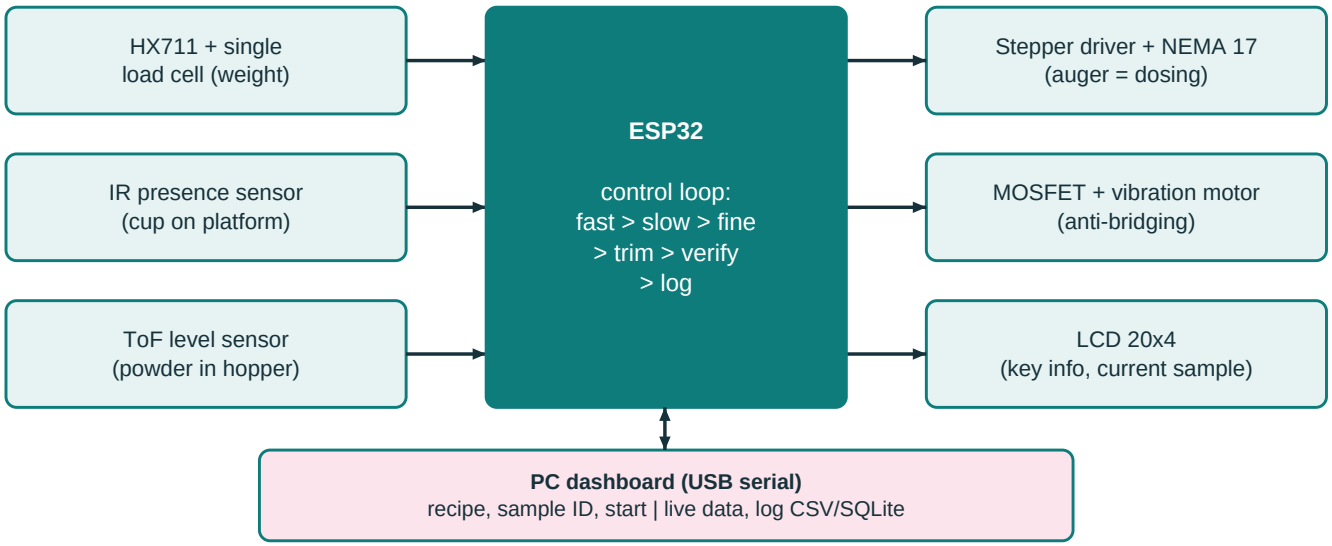

Figure 1 - Block diagram. Arrows show the direction of data and commands: sensors feed the ESP32; the ESP32 drives the actuators and the display; the PC exchanges recipe, commands and logs with the ESP32.

4. What will the minimum demo show after the 18 hours?

A live run of a real recipe, covering the full loop (instructions in, physical action, monitoring, verification, digital record):

- the user enters a target (for example 10.0 g) and a tolerance on the PC, together with a sample ID;
- the start checks turn green, then the auger doses in **three stages driven by the weight reading**: fast, slow, then fine, followed by a trim loop;
- the LCD and the PC show **target, measured value, deviation and Accepted/Rejected**, and one log line is written;
- **error handling live:** removing the cup pauses the run and flags an error; an auger that turns while the weight does not rise is flagged as "no flow";
- **measured gain:** a series of doses (target 10 runs) compared with manual spatula dosing: mean error, spread, rejected doses and time per dose.

Bonus, only if time allows: overshoot compensation that learns from previous runs, a dashboard with run history and repeatability statistics, and a second identical hopper + auger module (one auger per material, no cross-contamination) for a two-line recipe.

5. How does the system monitor itself, verify its result, and log each operation?

Closed-loop dosing (weight-driven, never timed). Speed depends on the remaining mass. Powder can be added but not removed, so the control always approaches the target from below:

- **Fast** until about 75% of the target, **slow** until about 95%, then **fine** with a few microsteps at a time;
- the motor stops at a **stop threshold in grams** (target minus the expected overshoot, i.e. powder still falling after the stop), then the system waits about 0.5 s for the weight to settle;
- a **trim loop** adds 1 to 3 microsteps, waits and measures again, until the weight is within tolerance or a maximum number of attempts is reached;
- the load cell reading is averaged over several samples, the tare is repeated before each dose, and the overshoot and grams-per-step values are updated from previous runs (moving average).

Self-monitoring. The system checks cup presence (IR sensor plus weight), tare stability, hopper level (ToF sensor), no flow (auger turning, weight not rising), excessive or too slow dosing (timeout), and cup removed during a dose. Any abnormal case stops the motor and is flagged.

Verification. After settling, measured mass is compared with the target. Status is **Accepted** if the deviation is within the tolerance (chosen by the team, for example 0.1 g), **Rejected** if it is outside, and **Incomplete/Error** if the run failed. Example: Target 10.0 g, Measured 10.1 g, Deviation +0.1 g, Accepted.

Logging. Each dose writes one record on the PC (CSV/SQLite): recipe ID, material ID, batch/sample ID, target, measured, deviation, timestamps, number of trim steps, overshoot, error message and final status. Rejected and failed runs are kept, never overwritten. The ESP32 also shows the last result on the LCD, so a result remains visible if the PC link is lost.